

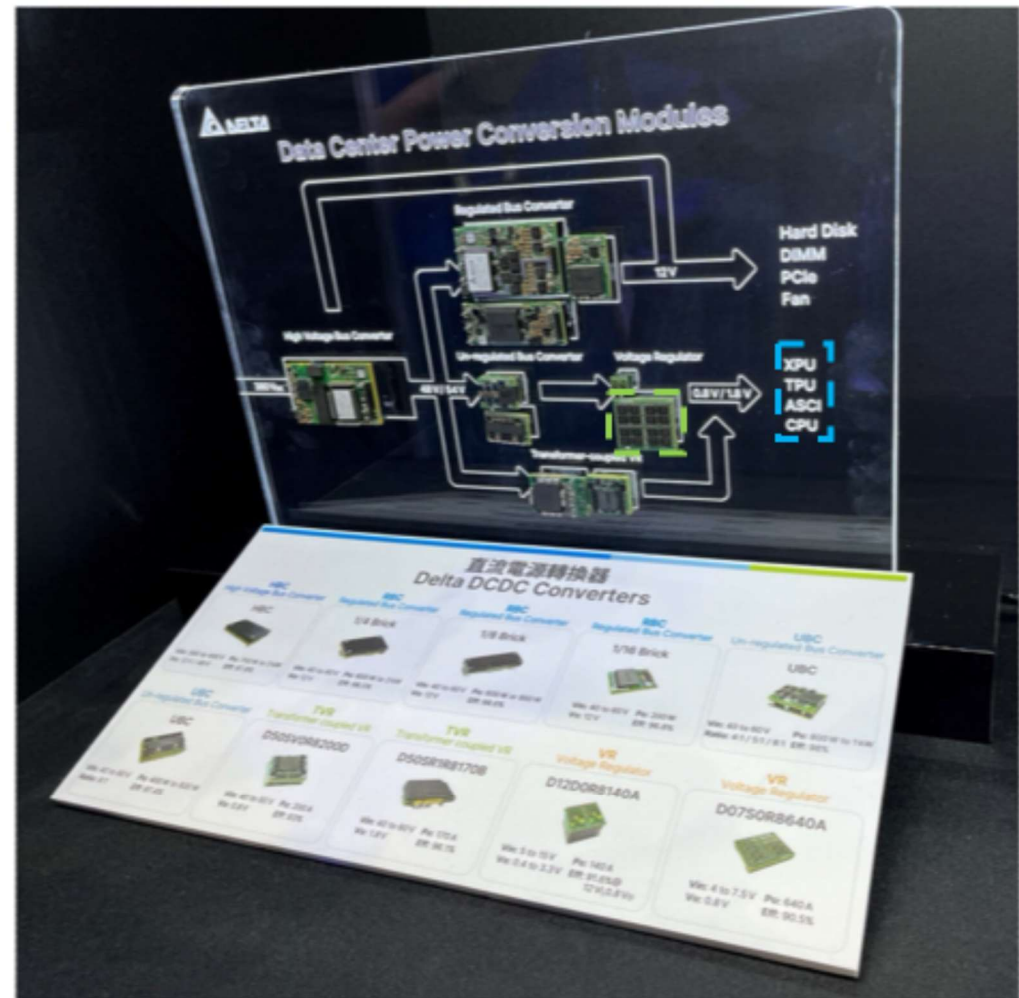
Exhibit 27

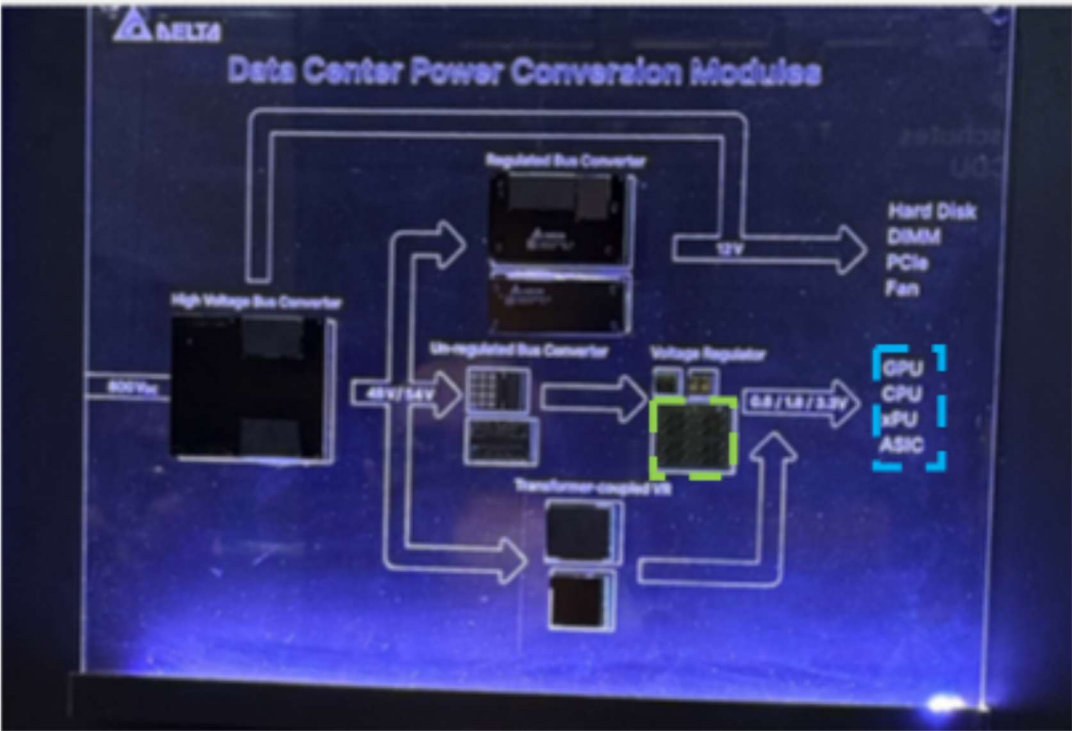
Claim Chart for the Practice of Independent Claims 1, 2, 40, 45, 49, 53, 67, and 68 of Complainant Vicor Corporation's United States Patent No. 10,903,734 by Delta VPD Module and Circuit Board Assemblies and Computing Systems Containing the Same¹

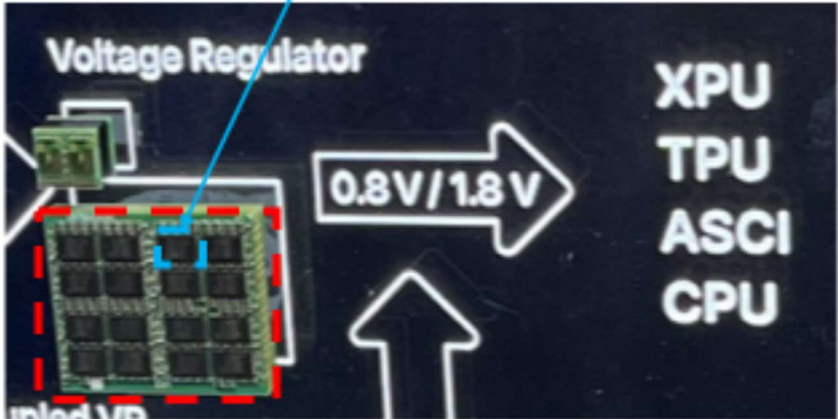
Claim 1

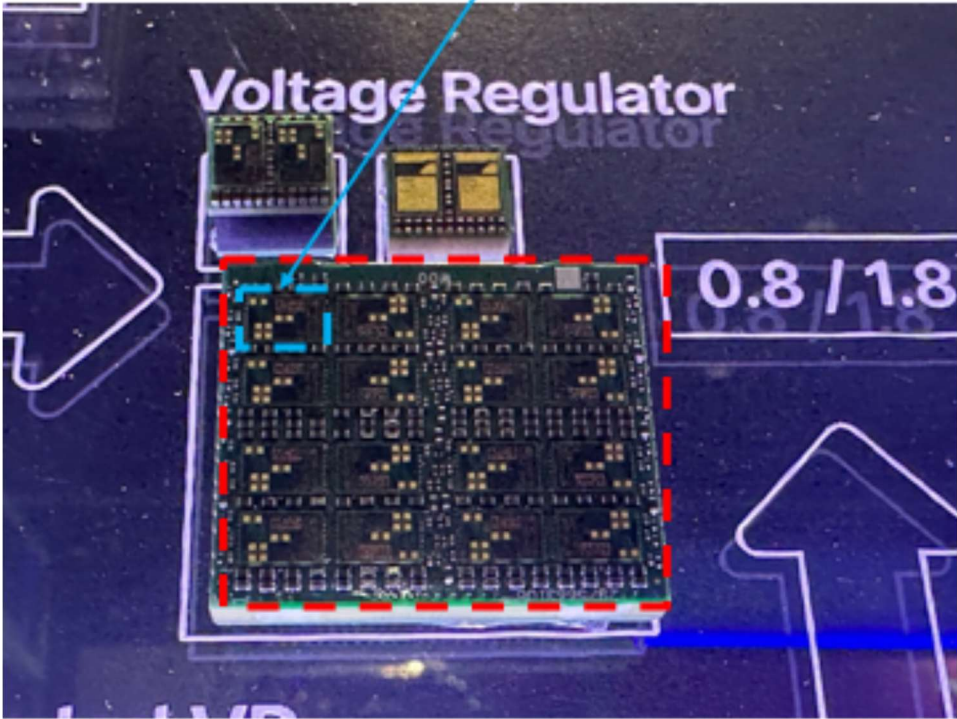
#	Limitation	Practice Evidence
1.pre	A method of powering a semiconductor device having power input terminals, the method comprising:	The Delta VPD Module [green] practices “[a] method of powering a semiconductor device [blue] having power input terminals.”

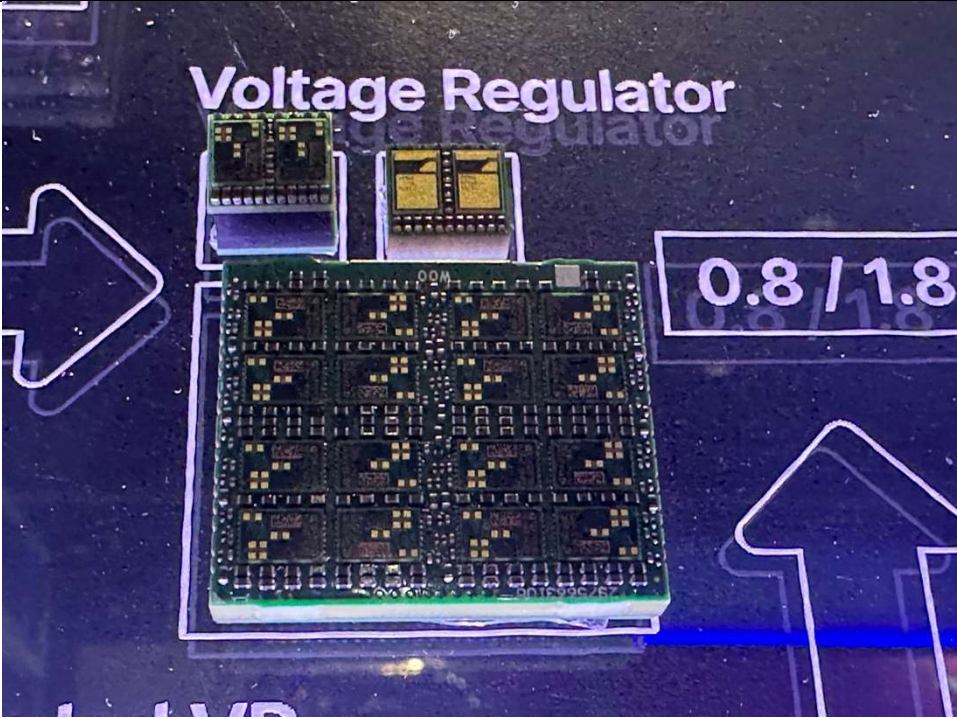
¹ The following analysis of Delta VPD Module is applicable to all other equivalent Delta power modules as well as to circuit board assemblies and computing systems containing the same.



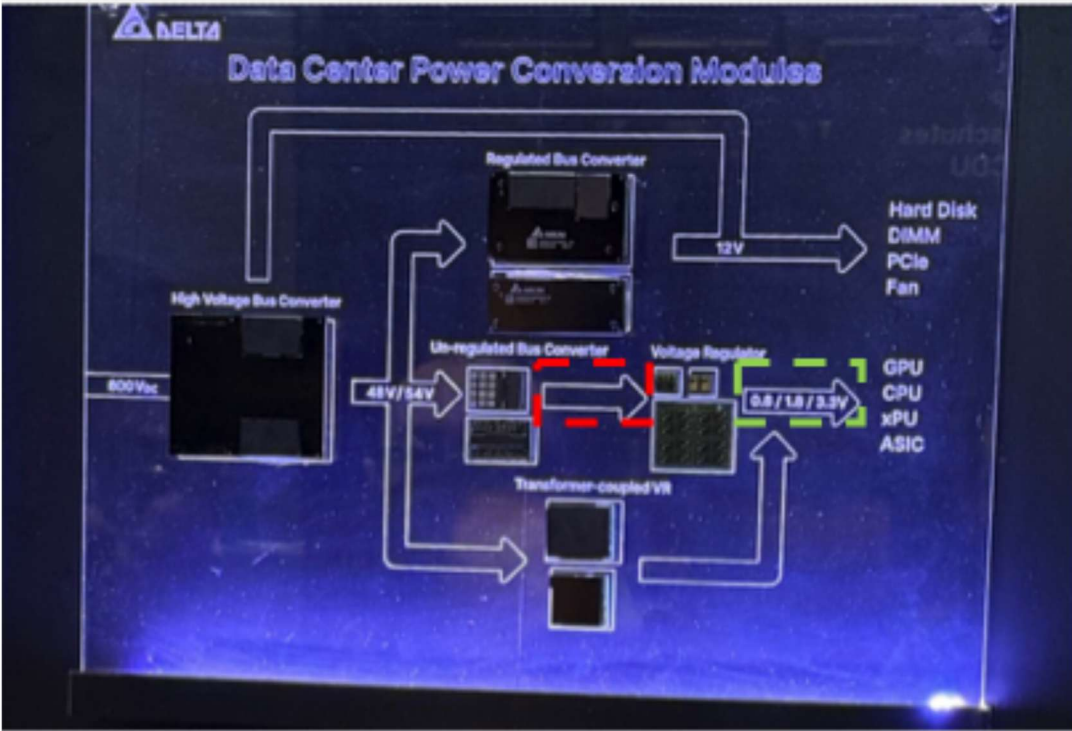
#	Limitation	Practice Evidence
		Ex. 28 at 1.  Ex. 28 at 2.
1.a	providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	The Delta VPD Module practices a method comprising “providing a power conversion module [red] having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells [blue].”

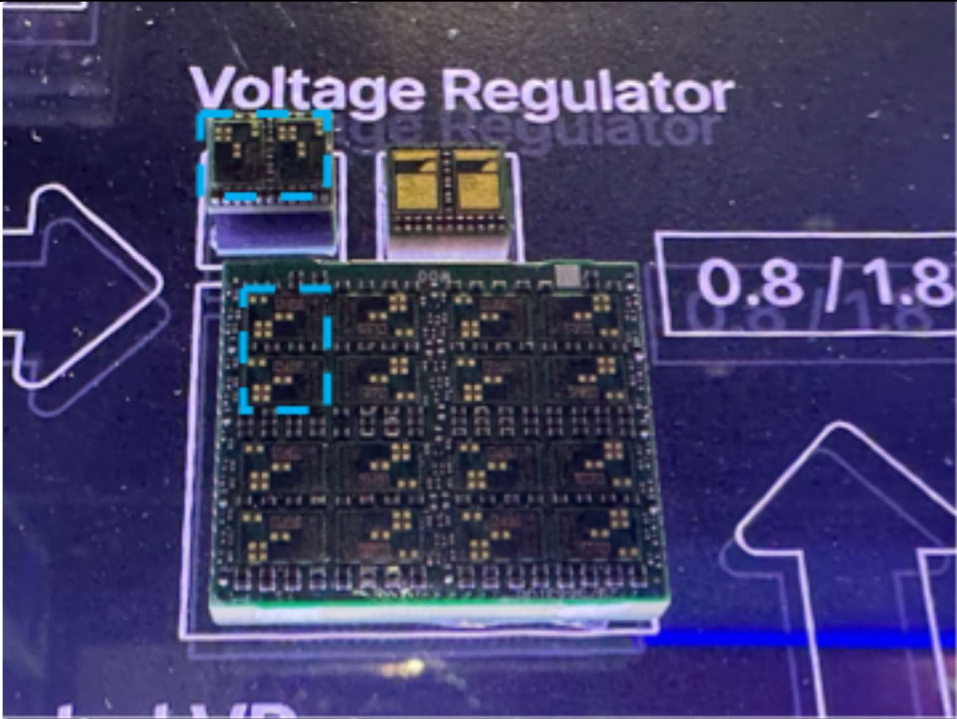
#	Limitation	Practice Evidence
		Example of a conversion cell  Ex. 28 at 1. The diagram illustrates a voltage regulation system. On the left, a green printed circuit board (PCB) is shown with a voltage regulator chip and several electrolytic capacitors. A red dashed rectangle highlights a specific section of the PCB. A blue arrow points from the text 'Example of a conversion cell' to this highlighted area. To the right of the PCB, a large white arrow points right, containing the text '0.8V/1.8V'. Below this arrow is another white arrow pointing up. To the far right, the text 'XPU', 'TPU', 'ASCI', and 'CPU' is listed vertically, representing the components that receive the regulated voltage.

#	Limitation	Practice Evidence
		Example of a conversion cell  Ex. 28 at 3.
1.b	each conversion cell including one or more converter output terminals,	The Delta VPD Module practices a method comprising “each conversion cell including one or more converter output terminals.” Each conversion cell (see 1.a) includes a positive output terminal (Vout) and a ground output terminal (GND).

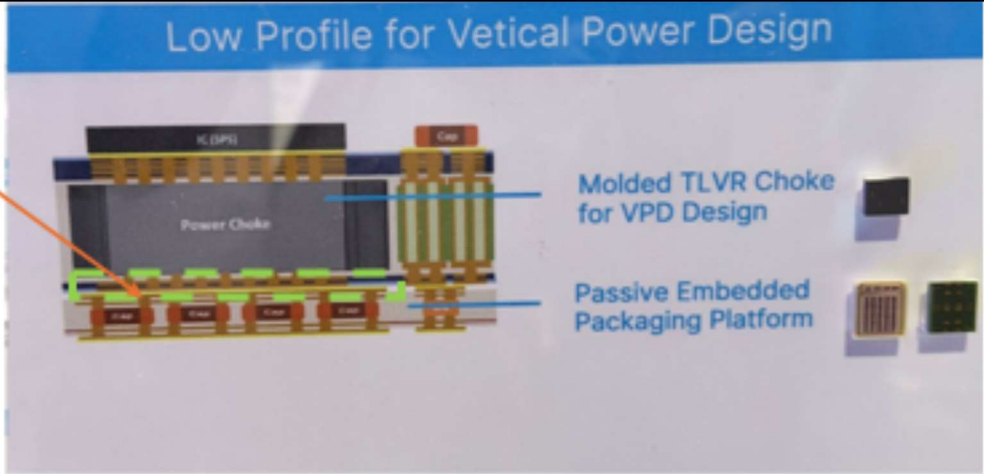
#	Limitation	Practice Evidence
		 Ex. 28 at 3. <i>See also</i> 1.a.
1.c	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	The Delta VPD Module practices a method comprising “the conversion circuitry being configured to convert power received at a first voltage [red] for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage [green].” The un-regulated bus converters down-converts voltage at either 4:1, 5:1, or 8:1 [purple]. Thus the “first voltage” is at least 6V (i.e., 48V / 8), which is higher than the “second lower voltage” of either 0.8V or 1.8V.

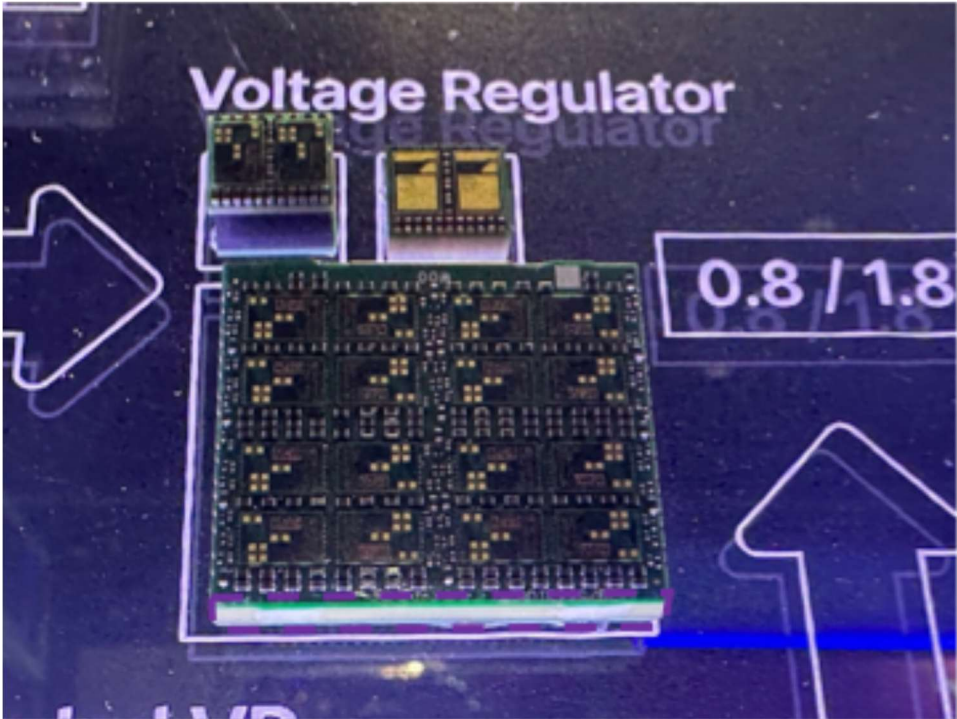


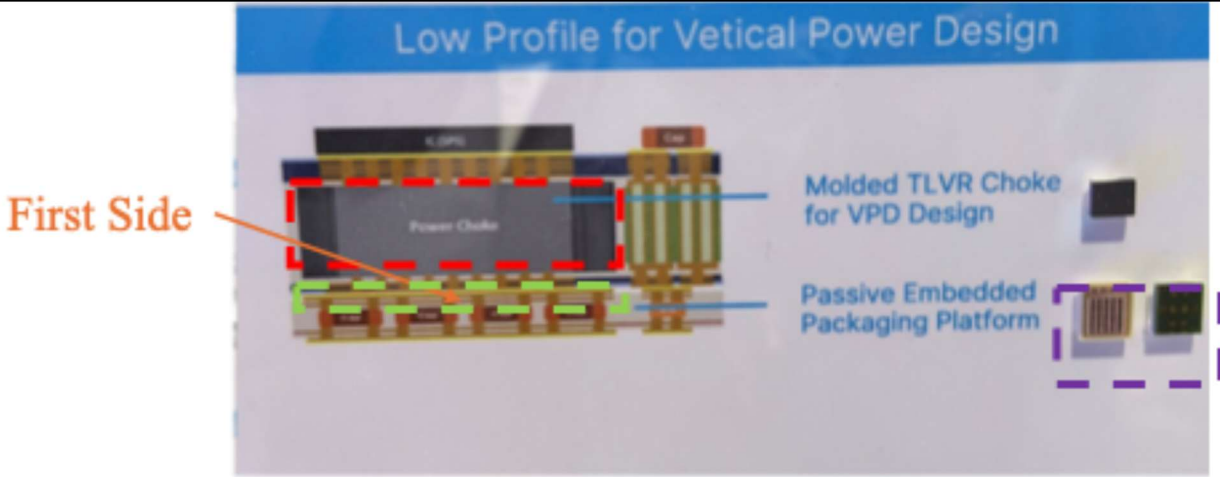
#	Limitation	Practice Evidence
		Ex. 28 at 1.  Ex. 28 at 2.
1.d	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals	The Delta VPD Module practices a method comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” The converter output terminals are arranged in a first pattern at a first spacing greater than 1mm corresponding to each Voltage Regulator conversion cell

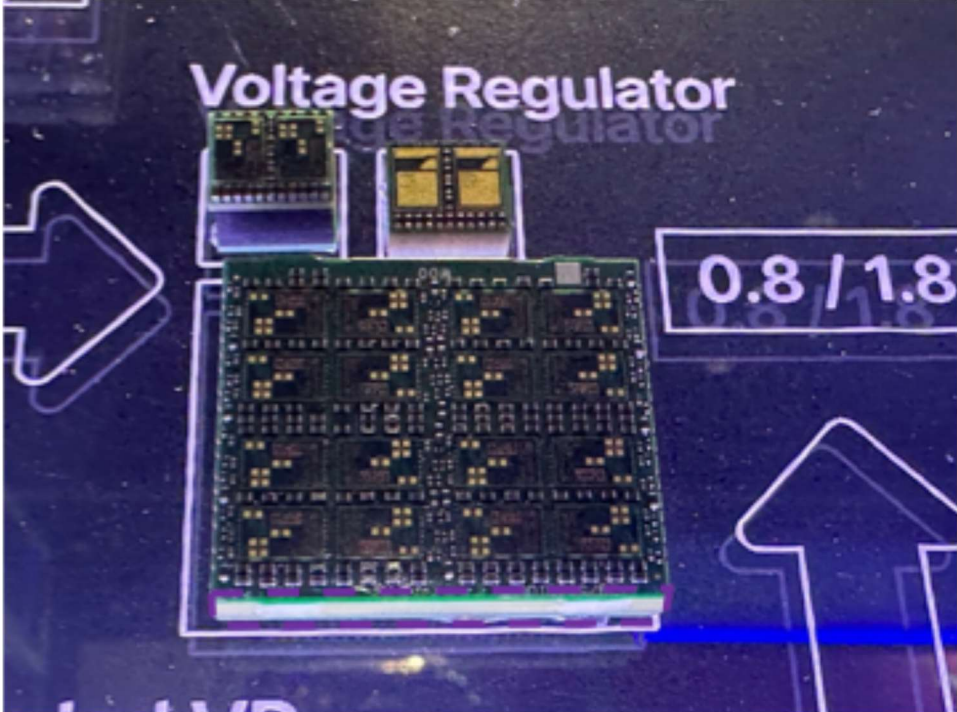
#	Limitation	Practice Evidence
		 Ex. 28 at 3.
1.e	providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	The Delta VPD Module practices a method comprising “providing an interconnection module [purple] having x-input terminals [green] arranged on a first side [orange] of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.”

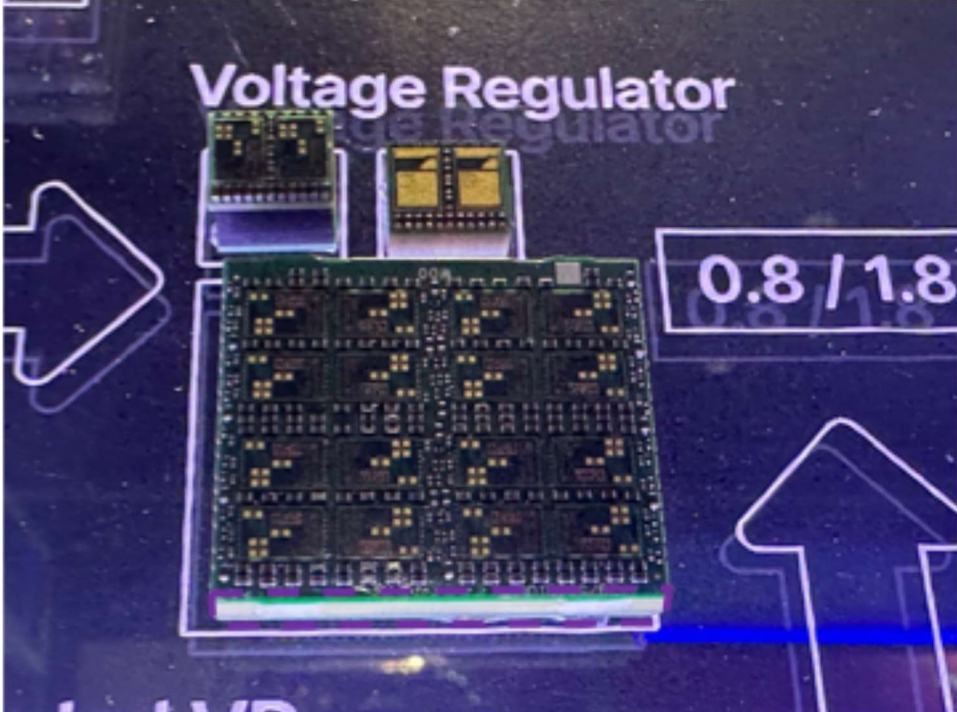
#	Limitation	Practice Evidence
		Ex. 28 at 3.

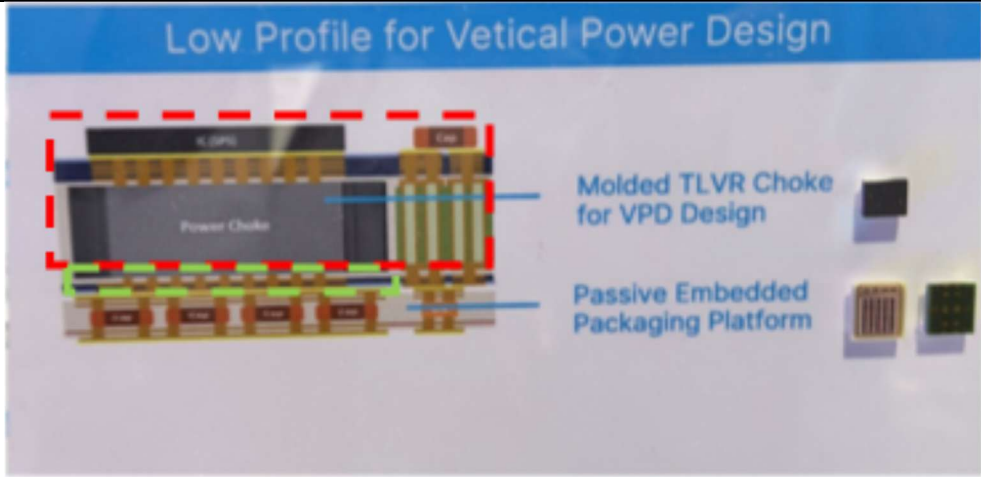
#	Limitation	Practice Evidence
		First Side  Ex. 28 at 4.
1.f	x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	The Delta VPD Module practices a method comprising “x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” Delta’s interconnection module’s second side contains a grid of output terminals configured to align with the power input terminals of the semiconductor device (or ASIC) and arranged in a second pattern with a second spacing of 1 mm or less.
1.g	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	The Delta VPD Module practices a method comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.”

#	Limitation	Practice Evidence
		Delta's VPD module's second spacing of around 1 mm or less is smaller and, thus, different than the first spacing which is greater than around 1mm.
1.h	assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;	The Delta VPD Module practices a method comprising “assembling the power conversion module [red] to the first side [orange] of the interconnection module [purple] with each converter output terminal mated with and electrically connected to a respective x-input terminal.” Upon information and belief, Delta VPD's converter output terminals are mated with and electrically connected to a respective x-input terminal of the interconnection module, depicted below.  Ex. 28 at 3.

#	Limitation	Practice Evidence
		 Ex. 28 at 4.
1.i	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	The Delta VPD Module practices a method comprising “wherein the interconnection module [purple] is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” Delta’s interconnection module satisfies this limitation as it routes current from the converter output terminals arranged in a first pattern at a first spacing greater than 1mm to the semiconductor device in the second pattern at a second spacing less than 1mm.

#	Limitation	Practice Evidence
		 Ex. 28 at 3.
1.j	wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and	The Delta VPD Module practices a method comprising “wherein the power conversion module [red] and the interconnection module [purple] are stacked to form a module sandwich including the power conversion module [red] in a first layer, electrical connections between the converter output terminals and the x-input terminals [green] in a second layer, and the interconnection module [purple] in a third layer.”

#	Limitation	Practice Evidence
	the interconnection module in a third layer.	 Ex. 28 at 3.

#	Limitation	Practice Evidence
		 Ex. 28 at 4.

Claim 2

#	Limitation	Practice Evidence
2.pre	A method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged	The Delta VPD Module practices a method comprising “[a] method of powering a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die, the method comprising:”

#	Limitation	Practice Evidence
	within the footprint of the die, the method comprising:	Ex. 28 at 3. <i>See also</i> 1.pre, 1.i.
2.a	providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	The Delta VPD Module practices a method comprising “providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” <i>See</i> 1.a.

#	Limitation	Practice Evidence
2.b	each conversion cell including one or more converter output terminals,	The Delta VPD Module practices a method comprising “each conversion cell including one or more converter output terminals.” <i>See 1.b.</i>
2.c	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	The Delta VPD Module practices a method comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
2.d	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;	The Delta VPD Module practices a method comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” <i>See 1.d.</i>
2.e	providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	The Delta VPD Module practices a method comprising “providing an interconnection module having x-input terminals arranged on a first side of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.” <i>See 1.e.</i>
2.f	x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input	The Delta VPD Module practices a method comprising “x-output terminals arranged on a surface of a second opposite side of the interconnection module in a second pattern arranged within the die footprint and configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.”

#	Limitation	Practice Evidence
	terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	<i>See 1.f, 2.pre.</i>
2.g	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	The Delta VPD Module practices a method comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.” <i>See 1.g.</i>
2.h	assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal;	The Delta VPD Module practices a method comprising “assembling the power conversion module to the first side of the interconnection module with each converter output terminal mated with and electrically connected to a respective x-input terminal.” <i>See 1.h.</i>
2.i	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	The Delta VPD Module practices a method comprising “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” <i>See 1.i.</i>
2.j	wherein the power conversion module and the interconnection module are stacked to form a	The Delta VPD Module practices a method comprising “wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion

#	Limitation	Practice Evidence
	module sandwich including the power conversion module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.	module in a first layer, electrical connections between the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.” <i>See 1.j.</i>

Claim 40

#	Limitation	Practice Evidence
40.pre	An apparatus comprising:	Delta VPD Module is part of “[a]n apparatus.” <i>See 1.pre.</i>
40.a	a semiconductor device having power input terminals;	Delta VPD Module is part of an apparatus comprising “a semiconductor device having power input terminals.” <i>See 2.pre.</i>
40.b	a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	Delta VPD Module is part of an apparatus comprising “a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” <i>See 1.a.</i>
40.c	each conversion cell including one or more converter output terminals,	Delta VPD Module is part of an apparatus comprising “each conversion cell including one or more converter output terminals.”

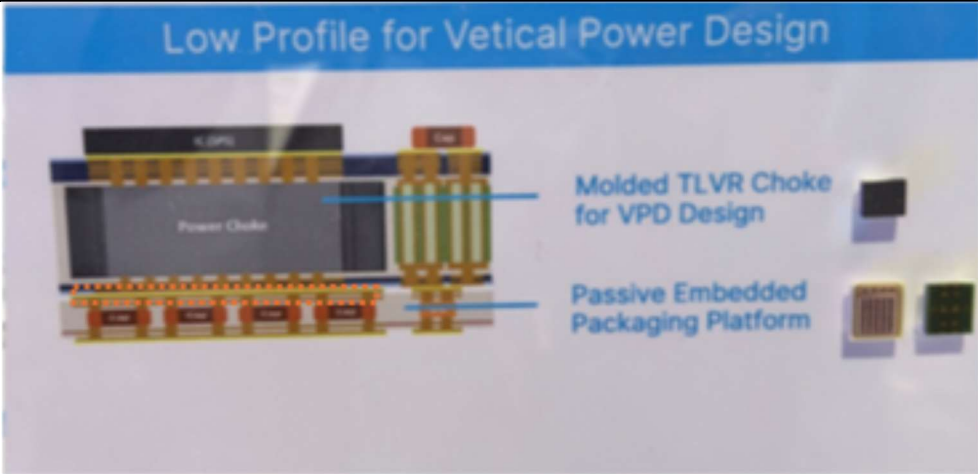
#	Limitation	Practice Evidence
		<i>See 1.b.</i>
40.d	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	Delta VPD Module is part of an apparatus comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
40.e	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and	Delta VPD Module is part of an apparatus comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” <i>See 1.d.</i>
40.f	an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	Delta VPD Module is part of an apparatus comprising “an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.” <i>See 1.e.</i>
40.g	x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	Delta VPD Module is part of an apparatus comprising “x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” <i>See 1.f.</i>
40.h	the second spacing being different than the first spacing, the x-input terminals being	Delta VPD Module is part of an apparatus comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.”

#	Limitation	Practice Evidence
	electrically connected to the x-output terminals;	<i>See 1.g.</i>
40.i	the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;	Delta VPD Module is part of an apparatus comprising “the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module.” <i>See 1.h.</i>
40.j	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	Delta VPD Module is part of an apparatus “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” <i>See 1.i.</i>
40.k	wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.	Delta VPD Module is part of an apparatus “wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.” <i>See 1.j.</i>

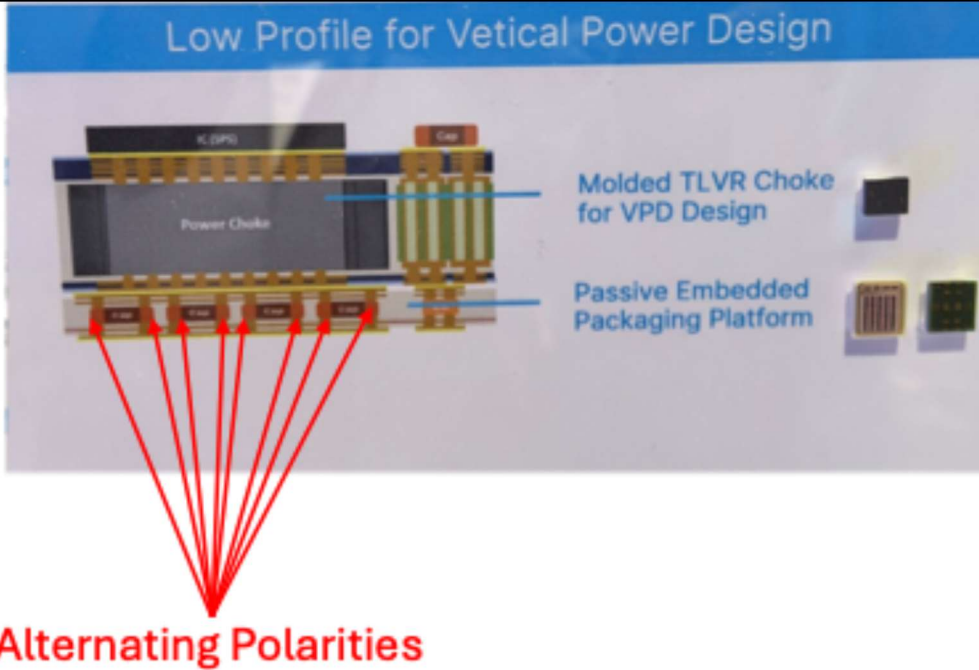
Claim 45

#	Limitation	Practice Evidence
45.pre	A method of powering a semiconductor device having power input terminals, the method comprising:	Delta VPD Module practices “[a] method of powering a semiconductor device having power input terminals.” <i>See 2.pre.</i>
45.a	providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	Delta VPD Module practices a method comprising “providing a power conversion module having conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” <i>See 1.a.</i>
45.b	each conversion cell including one or more converter output terminals,	Delta VPD Module practices a method comprising “each conversion cell including one or more converter output terminals.” <i>See 1.b.</i>
45.c	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	Delta VPD Module practices a method comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
45.d	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals;	Delta VPD Module practices a method comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.” <i>See 1.d.</i>
45.e	providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with	Delta VPD Module practices a method comprising “providing an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and”

#	Limitation	Practice Evidence
	the first pattern of converter output terminals, and	<i>See 1.e.</i>
45.f	x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	Delta VPD Module practices a method comprising “x-output terminals arranged on a second surface in a second pattern configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” <i>See 1.f.</i>
45.g	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	Delta VPD Module practices a method comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.” <i>See 1.g.</i>
45.h	providing a plurality of conductive pillars for electrically connecting the converter output terminals with respective x-input terminals;	Delta VPD Module practices a method comprising “providing a plurality of conductive pillars [orange] for electrically connecting the converter output terminals with respective x-input terminals.” <i>See 1.b</i>

#	Limitation	Practice Evidence
		 Ex. 28 at 4.
45.i	assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars;	Delta VPD Module practices a method comprising “assembling the power conversion module to the interconnection module with each converter output terminal aligned with and electrically connected to a respective x-input terminal by a respective one of the conductive pillars.” See 1.h, 45.h.
45.j	wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor	Delta VPD Module practices a method “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” See 1.i.

#	Limitation	Practice Evidence
	device arranged in the second pattern;	
45.k	wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer; and	Delta VPD Module practices a method “wherein the power conversion module and the interconnection module are stacked to form a module sandwich including the power conversion module in a first layer, the conductive pillars connecting the converter output terminals and the x-input terminals in a second layer, and the interconnection module in a third layer.” <i>See 1.j.</i>
45.l	wherein the x-output terminals are arranged with alternating polarities to reduce interconnection inductance.	Delta VPD Module practices a method “wherein the x-output terminals [red] are arranged with alternating polarities to reduce interconnection inductance.”

#	Limitation	Practice Evidence
		 Ex. 28 at 4.

Claim 49

#	Limitation	Practice Evidence
49.pre	An apparatus comprising:	Delta VPD Module is part of “[a]n apparatus.” <i>See</i> 1.pre.

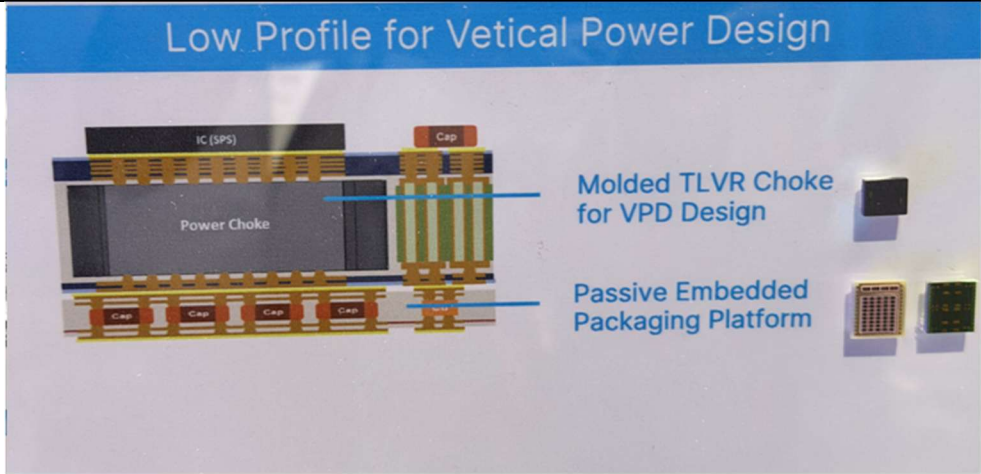
#	Limitation	Practice Evidence
49.a	a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die;	Delta VPD Module is part of an apparatus comprising “a semiconductor device, including at least one semiconductor die having a plurality of die power input terminals arranged on a surface of the die and having edges of the semiconductor die defining a die footprint, the semiconductor device having power input terminals arranged within the footprint of the die.” <i>See 2.pre.</i>
49.b	a power conversion module having switching power conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells,	Delta VPD Module is part of an apparatus comprising “a power conversion module having switching power conversion circuitry arranged in a multi-cell structure having a plurality of conversion cells.” <i>See 1.a.</i>
49.c	each conversion cell including one or more converter output terminals,	Delta VPD Module is part of an apparatus comprising “each conversion cell including one or more converter output terminals.” <i>See 1.b.</i>
49.d	the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage,	Delta VPD Module is part of an apparatus comprising “the conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals of the plurality of conversion cells at a second lower voltage.” <i>See 1.c.</i>
49.e	the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals; and	Delta VPD Module is part of an apparatus comprising “the converter output terminals being arranged in a first pattern having a first spacing between the converter output terminals.”

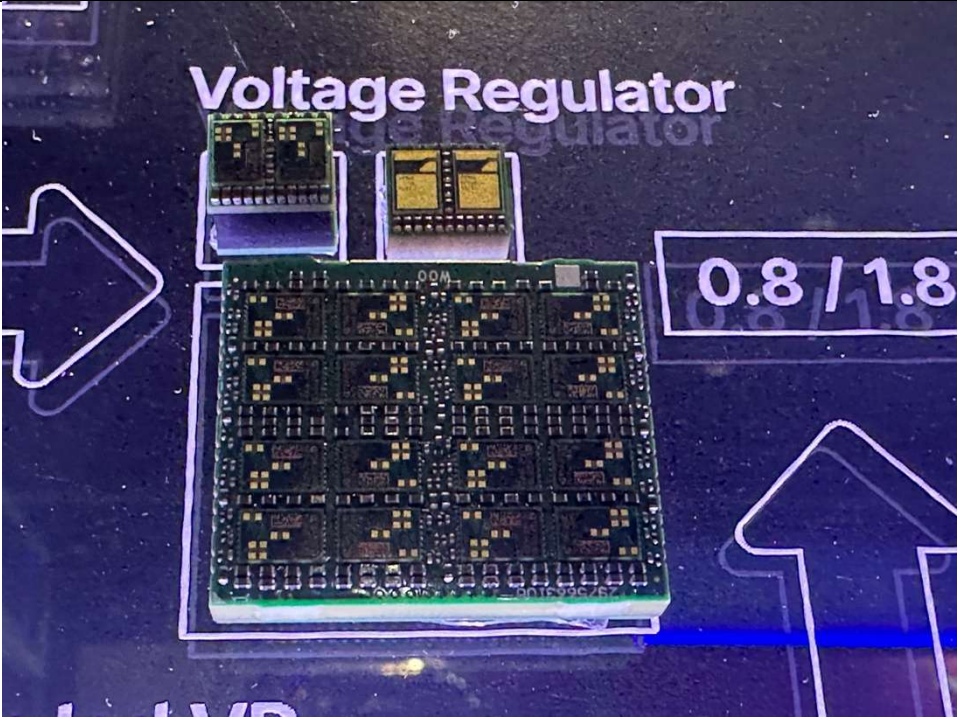
#	Limitation	Practice Evidence
		<i>See 1.d.</i>
49.f	an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals, and	Delta VPD Module is part of an apparatus comprising “an interconnection module having x-input terminals arranged on a first surface of the interconnection module in a pattern configured to align with the first pattern of converter output terminals.” <i>See 1.e.</i>
49.g	x-output terminals arranged on a second surface in a second pattern with alternating polarities configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals,	Delta VPD Module is part of an apparatus comprising “x-output terminals arranged on a second surface in a second pattern with alternating polarities configured to align with the power input terminals of the semiconductor device, the second pattern having a second spacing between x-output terminals.” <i>See 1.f; 45.l.</i>
49.h	the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals;	Delta VPD Module is part of an apparatus comprising “the second spacing being different than the first spacing, the x-input terminals being electrically connected to the x-output terminals.” <i>See 1.g.</i>
49.i	the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module;	Delta VPD Module is part of an apparatus comprising “the converter output terminals of the power conversion module being mated with and electrically connected to respective x-input terminals of the interconnection module.” <i>See 1.h.</i>
49.j	wherein the interconnection module is configured to provide electrical and mechanical	Delta VPD Module is part of an apparatus “wherein the interconnection module is configured to provide electrical and mechanical translation between the converter output terminals arranged in the

#	Limitation	Practice Evidence
	translation between the converter output terminals arranged in the first pattern and the power input terminals of the semiconductor device arranged in the second pattern;	first pattern and the power input terminals of the semiconductor device arranged in the second pattern.” <i>See 1.i.</i>
49.k	wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.	Delta VPD Module is part of an apparatus “wherein the power conversion module and the interconnection module are stacked with the power conversion module located at a first level, electrical connections between the converter output terminals and the x-input terminals at a second level, and the interconnection module at a third level.” <i>See 1.j.</i>

Claim 53

#	Limitation	Practice Evidence
53.pre	An apparatus for delivering power vertically to a semiconductor die, the die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint, the apparatus comprising:	Delta VPD Module is “[a]n apparatus for delivering power vertically to a semiconductor die, the die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint.” <i>See 2.pre.</i> Delta VPD Module is “[a]n apparatus for delivering power vertically to a semiconductor die.”

#	Limitation	Practice Evidence
		 Ex. 28 at 4.

#	Limitation	Practice Evidence
		 Ex. 28 at 3.
53.a	switching power conversion circuitry arranged on a first substrate having a plurality of switching power conversion cells and a plurality of converter output terminals,	Delta VPD Module is an apparatus comprising “switching power conversion circuitry arranged on a first substrate having a plurality of switching power conversion cells and a plurality of converter output terminals.” <i>See</i> 49.b, 1.a, 1.b.

#	Limitation	Practice Evidence
53.b	the switching power conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals at a second lower voltage;	Delta VPD Module is an apparatus comprising “the switching power conversion circuitry being configured to convert power received at a first voltage for delivery via the converter output terminals at a second lower voltage.” <i>See 1.c.</i>
53.c	translation interconnections including X-input terminals configured and arranged to align with and electrically connect to the converter output terminals,	Delta VPD Module is an apparatus comprising “translation interconnections including X-input terminals configured and arranged to align with and electrically connect to the converter output terminals.” <i>See 1.e.</i>
53.d	a plurality of X-output terminals configured and arranged with alternating polarities to align with respective power input terminals of the semiconductor die, and	Delta VPD Module is an apparatus comprising “a plurality of X-output terminals configured and arranged with alternating polarities to align with respective power input terminals of the semiconductor die.” <i>See 1.f, 45.1.</i>
53.e	a plurality of vertical conductors to make electrical connections to the X-output terminals;	Delta VPD Module is an apparatus comprising “a plurality of vertical conductors to make electrical connections to the X-output terminals [orange].”

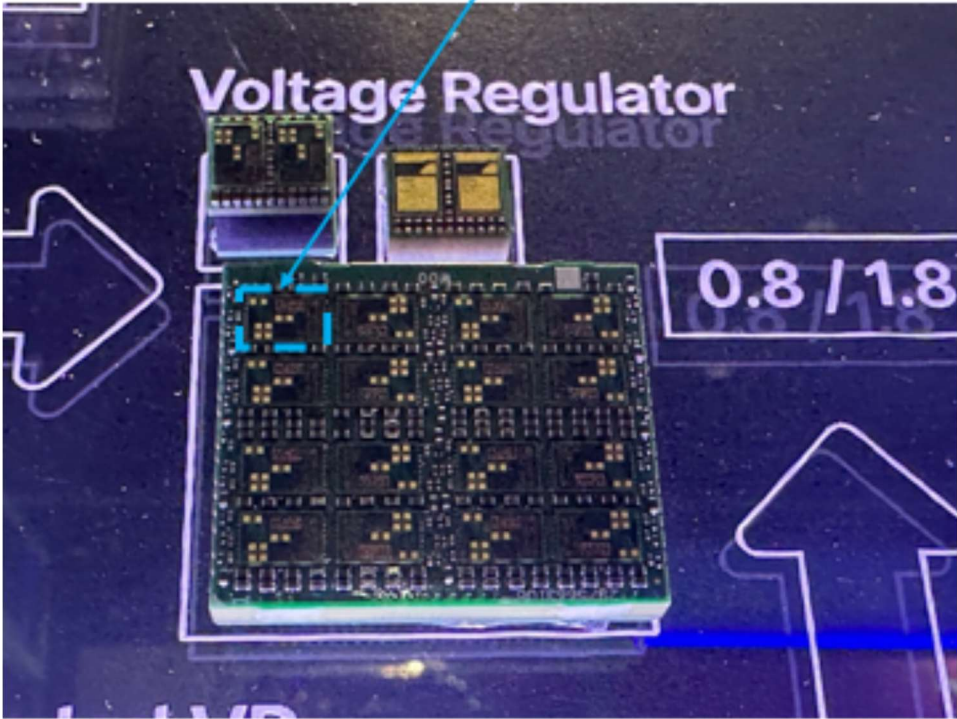
#	Limitation	Practice Evidence
		Ex. 28 at 4.
53.f	a capacitive array arranged between the power conversion circuitry and the semiconductor die and electrically connected to the translation interconnections to provide energy storage;	Delta VPD Module is an apparatus comprising “a capacitive array [blue] arranged between the power conversion circuitry [red] and the semiconductor die and electrically connected to the translation interconnections to provide energy storage.”

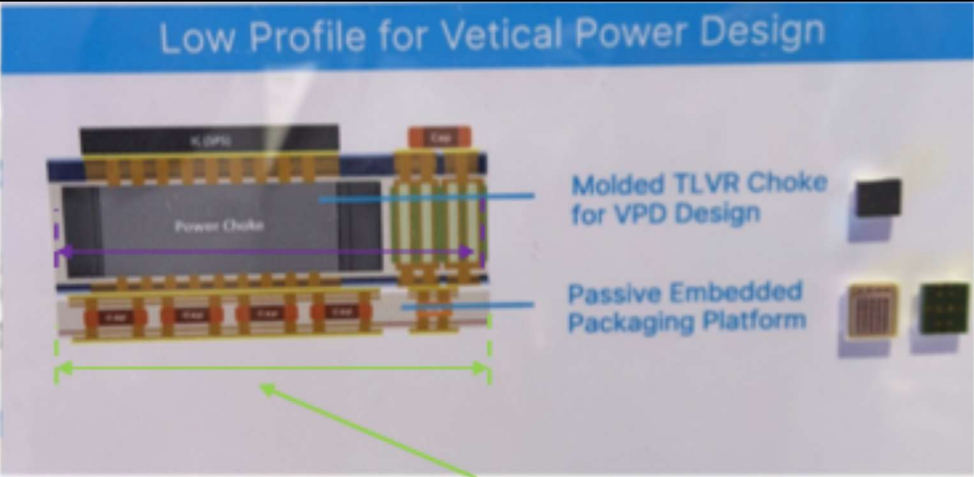
#	Limitation	Practice Evidence
		Ex. 28 at 4. <i>See</i> 1.i.
53.g	wherein the translation interconnections are configured to provide electrical and mechanical translation between the converter output terminals and the power input terminals of the semiconductor die;	Delta VPD Module is an apparatus “wherein the translation interconnections are configured to provide electrical and mechanical translation between the converter output terminals and the power input terminals of the semiconductor die.” <i>See</i> 1.i.
53.h	wherein the switching power conversion circuitry, the translation interconnections, and the capacitive array, are each configured and arranged within a respective footprint configured to occupy substantially all of the die	Delta VPD Module is an apparatus “wherein the switching power conversion circuitry, the translation interconnections, and the capacitive array, are each configured and arranged within a respective footprint configured to occupy substantially all of the die footprint, and are together configured and arranged to form a vertical stack configured to occupy substantially all of the die footprint.” <i>See</i> 2.pre, 53.f.

#	Limitation	Practice Evidence
	footprint, and are together configured and arranged to form a vertical stack configured to occupy substantially all of the die footprint.	

Claim 67

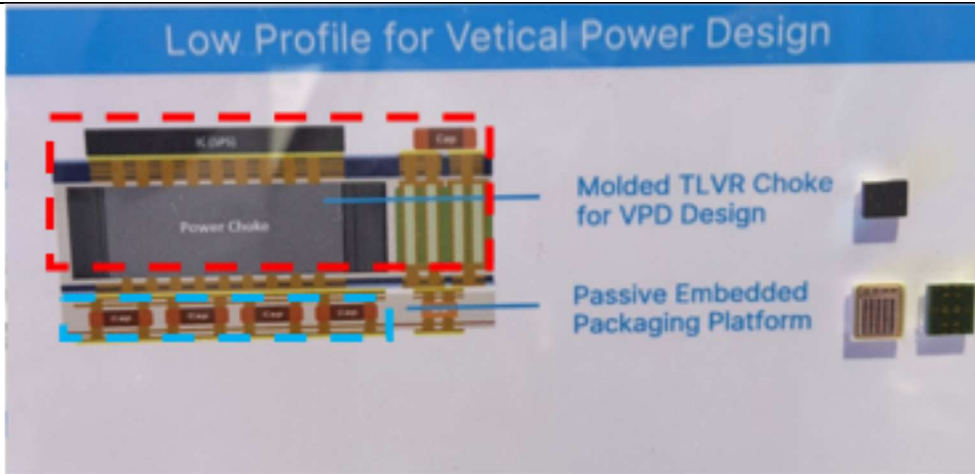
#	Limitation	Practice Evidence
67.pre	A vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint;	Delta VPD Module is “[a] vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint.” <i>See 53.pre.</i>
67.a	the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising:	Delta VPD Module is “the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint [green].” <i>See 2.pre.</i>
67.b	(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint, (ii) including a plurality of switching power conversion cells connected in parallel to	Delta VPD Module is an apparatus comprising “(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint [purple], (ii) including a plurality of switching power conversion cells [blue] connected in parallel to convert power from an input at an input voltage for delivery to the semiconductor power input terminals at a die voltage.”

#	Limitation	Practice Evidence
	convert power from an input at an input voltage for delivery to the semiconductor power input terminals at a die voltage,	Example of a conversion cell  Ex. 28 at 3.

#	Limitation	Practice Evidence
		 The minimal die footprint to accommodate connections Ex. 28 at 4. <i>See also</i> 1.c, 49.b, 2.pre.
67.c	(iii) the input voltage being at least five times higher than the die voltage, and (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint;	Delta VPD Module is an apparatus comprising “(iii) the input voltage being at least five times higher than the die voltage, and (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint.” The un-regulated bus converters down-converts voltage at either 4:1, 5:1, or 8:1 [purple]. Thus the “input voltage” is at least 6V (i.e., 48V / 8), which is “at least five times higher” than the 0.8V. Similarly, any voltage above 9V at the input of the voltage regulator is “at least five times higher” than the 1.8V (e.g., 48V/4, 48V/5, 54V/4, 54V/5)



#	Limitation	Practice Evidence
		Ex. 28 at 1. <i>See</i> 67.b.
67.d	(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint; and	Delta VPD Module is an apparatus comprising “(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint.” <i>See</i> 1.i, 45.1, 2.pre.
67.e	(c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals.	Delta VPD Module is an apparatus comprising “(c) a capacitive array [blue] (i) arranged between the power conversion circuitry [red] and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint and (iii) electrically connected to the interdigitated output terminals.”

#	Limitation	Practice Evidence
		 Ex. 28 at 4. See 45.1, 53.f, 53.g, 2.pre.

Claim 68

#	Limitation	Practice Evidence
68.pre	A vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint;	Delta VPD Module comprises “[a] vertical power delivery apparatus for a semiconductor die, the semiconductor die having a plurality of semiconductor power input terminals arranged on a surface of the semiconductor die and having edges of the semiconductor die defining a die footprint.” See 67.pre.

#	Limitation	Practice Evidence
68.a	the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint, the vertical stack comprising:	Delta VPD Module comprises “the vertical power delivery apparatus comprising in a vertical stack configured to occupy, in whole or in part, the die footprint.” <i>See 67.a.</i>
68.b	(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint, (ii) including a plurality of switching power conversion cells connected in parallel to convert power from an input at an input voltage to the semiconductor power input terminals at a die voltage,	Delta VPD Module is an apparatus comprising “(a) point-of-load (“POL”) switching power conversion circuitry (i) arranged on a substrate within a conversion footprint, (ii) including a plurality of switching power conversion cells connected in parallel to convert power from an input at an input voltage to the semiconductor power input terminals at a die voltage.” <i>See 67.b.</i>
68.c	(iii) the input voltage being at least five times higher than the die voltage, (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint;	Delta VPD Module is an apparatus comprising “(iii) the input voltage being at least five times higher than the die voltage, (iv) the conversion footprint being configured to occupy, in whole or in part, the die footprint.” <i>See 67.c.</i>
68.d	(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to	Delta VPD Module is an apparatus comprising “(b) interdigitated output terminals (i) electrically connected to receive power from the switching power conversion cells, (ii) configured within a terminal footprint to match the plurality of semiconductor power input terminals, (iii) the terminal footprint being configured to occupy, in whole or in part, the die footprint.” <i>See 67.d.</i>

#	Limitation	Practice Evidence
	occupy, in whole or in part, the die footprint;	
68.e	(c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint, and (iii) electrically connected to the interdigitated output terminals; and	Delta VPD Module is an apparatus comprising “(c) a capacitive array (i) arranged between the power conversion circuitry and the semiconductor die, (ii) within a filter footprint configured to occupy, in whole or in part, the die footprint, and (iii) electrically connected to the interdigitated output terminals.” <i>See 68.e.</i>
68.f	(d) a plurality of vertical conductors arranged within the die footprint to connect the interdigitated output terminals to the switching power conversion cells.	Delta VPD Module is an apparatus comprising “(d) a plurality of vertical conductors arranged within the die footprint to connect the interdigitated output terminals to the switching power conversion cells.” <i>See 45.l, 53.c, 53.e, 53.h.</i>